

Replay Attack Detection in Automatic Speaker Verification using Transformer-based Audio Representations

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Abstract—Automatic Speaker Verification (ASV) systems are becoming an important aspect of modern authentication technology. Nevertheless, ASV systems are still often vulnerable to spoofing attacks, including replay attacks, in which an attacker plays back a recording of the authentic user's voice to convince the ASV system that they are that legitimate user. Countermeasures have been developed for detecting replay attacks based on Constant-Q Cepstral Coefficients (CQCC) and corresponding Gaussian Mixture Models (GMM), but those have only modest success because they don't generalize across different recording conditions. This paper proposes a Transformer-based Replay Attack Detection System that will use Wav2Vec2.0 and ECAPA-TDNN embeddings to detect replayed audio and ultimately spoofed audio. The model proposed in this paper uses self-supervised feature learning to capture the temporal and spectral relationships contained in the speech signal to enhance detection. Evaluated the proposed detection model using ASVspoof2019 and ASVspoof2021 datasets, and it significantly reduces the Equal Error Rate (EER) in comparison to CQCC-GMM. Ultimately, results show that the proposed model is robust and scalable to deploy a robust replay detection model for future generation of voice authentication systems. The proposed TRADS Model provides an accuracy rate of 97.2%, an Equal Error Rate of 2.1%.

Keywords: *Speaker Verification, Reply attack, ECAPA-TDNN, Spoofing, CQCC*

I. Introduction

Voice authentication systems changed how verify identity. They enable natural and touch-free methods for verifying a speaker's identity. Although there are advantages to voice authentication systems, the reliability of these systems may be undermined by spoofing techniques, namely replay attacks, which are easy to implement[1][2][3]. In a replay attack, a bad actor captures a genuine user's voice when they have a valid opportunity and subsequently plays the recording[4]. Replay attacks are difficult to detect because the played recording will have the same spectral signature as genuine speech[5][6][7].

Previous research on replay attack detection was mostly based on engineered acoustic features, such as

Constant-Q Cepstral Coefficients (CQCC), which are designed to resolve in time-frequency to separate natural speech from recorded speech. These features, in combination with Gaussian Mixture Model (GMM) classifiers are the basis for anti-spoofing systems and generated the features used in applications such as the ASVspoof2017 challenge[8][9]. Previous research produced substantial accuracy in controlled settings, although it suffered significant performance loss when there was channel variation, background noise, or differences in reproducing and capturing playback devices. The limiting factor is that the previous methods rely on pre-designed features which cannot adapt to new attack types never seen before[10][11].

As deep learning has advanced rapidly, researchers have focused on data-driven approaches to strengthen the limitations of traditional methods. Convolutional Neural Networks (CNNs) showed significant promise in the task of automatically learning discriminative features from spectrograms and discovering replay-specific distortions. Nonetheless, CNNs tends to learn only local dependencies, which reduces their effectiveness in capturing long-lasting temporal dynamics in speech. Consequently, models using audio, attention-based, and self-supervised learning were developed to mark a significant shift in the advancement of research within the domain of voice anti-spoofing[12].

One of the most prominent advancements in this vein was Wav2Vec2.0, a self-supervised architecture that learns contextualized representations of speech from large amounts of unlabeled in-domain data, and provides the framework to learn a rich and generalizable feature representation from audio data, which may capture subtle changes associated with voices[13]. Additionally, the ECAPA-TDNN network introduced an improved time-delay neural network architecture that focuses on channel attention and channel aggregation, leading to high-quality discriminative speaker embeddings. Recently, approaches that are hybrid models, consisting of combining the use of self-supervised embeddings,

have emerged as a way to better generalize in a variety of spoofing scenarios.

Building on these advancements, the system proposed integrates both Wav2Vec2.0 and ECAPA-TDNN embeddings into a transform-based classification architecture [14]. The self-attention mechanisms of the transformer encoder permit the model to learn global temporal relations, helping it to separate authentic speech patterns from artifacts caused by replay. The proposed hybrid solution combines the strengths of self-supervised feature extraction with transformer attention modeling to produce a hybrid and accurate replay attack detection system, worthy of real-world deployment.

II. RELATED WORK

Research into Replay Attack Detection has gone through phases from early acoustic signal processing techniques to the use of currently available deep learning and transformer-based methodologies. The initial studies used mainly engineered acoustic features (e.g., Constant-Q Cepstral Coefficients - CQCCs) as a discriminative method for spoof detection [1], which applied frequency information from the audio spectrogram through frequency domain analysis. CQCCs became the preferred feature of all early spoofing methods and were typically combined with Gaussian Mixture Models (GMMs) for identification purposes [2]. Handcrafted systems using CQCCs successfully distinguished between actual and spoofed audio samples when the tests were conducted within power-controlled environments, however they struggled to generalize across the varying playback devices, microphones, and environments, resulting in the inability to effectively adapt to different recording conditions.

The introduction of deep learning into the field of anti-spoofing has revolutionized the area of research by removing the necessity for engineered feature development. Convolutional Neural Networks (CNNs) were one of the earliest models to automatically extract feature patterns from replayed spectral representations of audio recordings (spectrograms) as demonstrated by Lavrendryeva and his colleagues [3]. CNNs learned and generalised distortion patterns caused by replaying a sample directly from the training set and performed significantly better than existing CQCC/GMMs. However, because of the way CNNs are structured (i.e. they operate on local spatial characteristics), they are unable to include long term contextualised information thus making them susceptible to replay.

A major progression in feature representation has been seen with the implementation of self-supervised learning techniques like Wav2Vec2.0 [4]. In contrast with supervised models, Wav2Vec2.0 derives its effective speech embeddings from unlabeled audio through a masked prediction objective, allowing

Wav2Vec2.0 to capture the many complex temporal dependencies, the phonetic make-up and the broader dynamic aspects of speech. Contextual embeddings provided by Wav2Vec2.0 have greatly improved the separation between real speech and replayed audio. At the same time, ECAPA-TDNN has developed a novel speaker embedding framework that leverages attention mechanisms at the channel level and represents time based events, resulting in a very stable speaker representation for the tasks of speaker verification and the detection of spoofing.

The recent literature indicates that hybrid learning models, combining self-supervised stem embeddings with attention weighted models, have shown strong results. Hybrid feature combination techniques introduced by [6] have been shown to greatly improve the generalization capabilities to the many different and unseen spoof conditions. Additionally, with the advent of transformer architectures, there has been significant strides made in understanding the way that they utilize a self-attention model to more effectively model both short and long-term dependencies in addition to distinguishing between real and replayed speech artifacts. Transformer encoders that bring together Wav2Vec2.0 and ECAPA-TDNN embeddings have been shown to have significantly greater performance than their counterparts utilizing CNN's or LSTM's (i.e. greater levels of robustness and accuracy) under spoof.

Contemporary replay detection research uses combinations of hybrid self-supervised learning and attention-based frameworks as the primary focus. The use of the discriminative capabilities of Wav2Vec2.0 in combination with the temporal sensitivity of ECAPA-TDNN and the contextual modeling provided by transformers aims to bring the performance of new methods closer to that of traditional handcrafted feature-based methods and state-of-the-art deep learning methods combining all three will set a new standard for detecting replay attacks in automated speaker verification systems, enabling them to deal with more complex and diversified types of spoofing in the current environment.

III. DATASET DESCRIPTION

The evaluation of the replay attack detection system was conducted on two commonly referenced datasets in the anti-spoofing literature, namely the ASVspoof2019 and ASVspoof2021 datasets. The ASVspoof2019 Physical Access dataset contains both original and replayed speech recordings created in various acoustic environments and for different playback scenarios. These original and replayed recordings were constructed to simulate realistic attack conditions by recording at various distances and by utilizing separate playback devices and microphones. In ASVspoof2021, the same concept was followed, but recordings were created using a

greater degree of diversity (multichannel recordings and various conditions), to create variability.

All recordings were resampled to a sample rate of 16 kHz and normalization of the amplitude. Before the playback attack simulation, the datasets were processed to remove noise, the silence was trimmed, and Voice Activity Detection (VAD) was applied to isolate only the speech. The datasets splitting was done to have training, validation, and testing datasets split in a ratio of roughly 70-15-15; the datasets were utilized as a basis for observed replay attacks and for the performance and generalization of the replay attack detection system.

IV. PROPOSED METHODOLOGY

The proposed method addresses the limitations of traditional handcrafted features based systems through the use of recent developments in self-supervised learning and transformer architecture. The system is divided into four main stages: Pre-processing, Feature Extraction, Hybrid Feature Fusion, and Classification with a transformer.

A. Reason for the Selected Methodology

This methodology was chosen because Wav2Vec2.0 substantiates deep contextual representations that can only be achieved via training from a huge repositories

of unlabeled audio; it helps refine some replay artifacts that handcrafted features leave out. Similarly,

ECAPA-TDNN captures highly discriminative speaker embeddings as a result of effective temporal modeling. For the same reason, transformer provide global temporal attention that facilitates observation of distortion in longer segment of speech. The sequential attention from Wav2Vec2.0, ECAPA-TDNN, with a transformer provides the system ability to capture fine detailed spectral characteristics and long-range dependencies in the same representation resulting in the representational features providing maximal robustness and accuracy on downstream tasks.

B. Detailed Architecture

The pre-processing portion normalizes the input speech signal. Next the system will remove any silent segments. Wav2Vec2.0 converts the cleaned waveform input into contextual embeddings, which reinforces the representation with detailed phonetic and temporal information. ECAPA-TDNN yields speaker dependent and channel sensitive embeddings, that will help distinguish replay features. The two embeddings are fused via maximum value into a hybrid feature or synthesised feature vector.

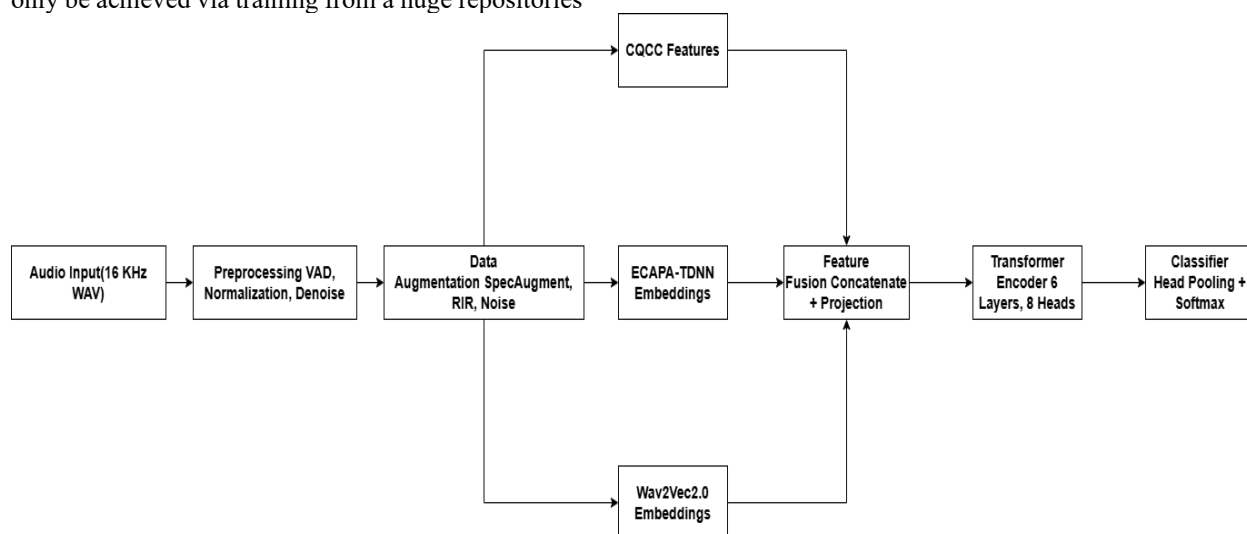


Fig 1: Proposed System Architecture

The proposed system is shown in Figure 1 and it integrates the results of three different speech feature extraction methods (CQCC, ECAPA-TDNN and Wav2Vec 2.0) into one output via the fusing module. The combined output of these 3 methods, which will form the basis for the multi-layer transformer encoder, along with the classifier, will allow the classification of real audio and audio that has been replayed.

The proposed work in this paper offers a very strong and modern architecture for detection in a replay attack scenario for Automatic Speaker Recognition

(ASV) systems, through the use of a combination of handcrafted, supervised, and self-supervised feature representations. The pipeline starts with a signal sampled at a 16 kHz rate, and it continues with a series of pre-processing steps: voice activity detection, normalising and denoising the signal. The purpose of these initial steps is to identify the relevant speech segments, and to reduce the amount of variation associated with background noise or inconsistent levels of recording, within the signal input to the ASV system. Various data augmentation techniques are applied to enhance the learning capability of the

model, including SpecAugment, simulated room impulse responses, and additive noise. These techniques augment the training process with examples of replayed speech in a variety of different replay environments and with a variety of different devices used to capture the original speaker.

A notable strength of this architecture lies in its use of three types of features extracted from three parallel branches. The CQCC branch captures the fine spectral distortions associated with replayed speech, the ECAPA-TDNN branch provides speaker discriminative and channel-aware representations, and finally, the Wav2Vec2.0 embeddings capture contextual and phonetic information from a self-supervised training of lavishing large audio corpora. The method used to create and fuse these heterogeneous features through concatenation and projection ensures that the ASV system takes full advantage of the complementary nature of these multiple sources of information; combining handcrafted, speaker-discriminative, and deep contextual features.

A multi-layered Transformer encoder, which uses eight separate heads for processing the inputs as well as the representation resulting from fusing, is used to process this fused representation. As such, it is able to capture the long-term temporal dependencies as well as fine-grained replay inconsistencies inherent in the data. The classifier head, which includes pooling and softmax, generates the spoofing probability. The entire design is supported by the understanding that it has a strong conceptual background and is in-line with current cutting-edge research into Spoof Detecting System. In addition, the hybrid, multi-branch design improves its robustness to different acoustic conditions; therefore, the overall design is believed to be particularly applicable to designs used to provide ASV security in modern systems.

The combined representation is then forwarded to a transformer encoder, with a multi-head self-attention mechanism, allowing the system to learn global dependencies along the speech sequence. This assists in recognizing the distortion associated with playback sources, microphones, and reflections from the environment. This transformer output is then placed into fully connected layers passing to a Softmax classifier predicting if the input is genuine or replayed.

Training uses the Adam optimizer with a learning rate of $1e-4$ and early stopping to reduce the impact of overfitting. The model is optimized for minimizing cross-entropy loss.

V. EVALUATION METRICS AND RESULTS

The proposed Replay Attack Detection System, which is based on Transformer, was benchmarked against traditional handcrafted models and newly developed deep learning-based systems, with respect to accuracy, robustness, and real-time performance. The assessment of the system incorporated four important performance measures in speaker verification - Accuracy, Equal Error Rate (EER), Mean Average Precision (mAP@0.5), and Inference Speed (FPS). Accuracy is defined as the proportion of validated samples and indicates overall capability of the system. Equal Error Rate is defined as the point where the false acceptance rate equals the false rejection rate and presents a balanced view of reliability and robustness of the system. Mean Average Precision indicates how well the model is able to achieve and maintain consistent detection precision across varying levels of confidence on an IoU threshold of 0.5 and allows assessment of discriminative power of the classifier. Inference speed (in FPS) indicates system capability for real-time or embedded scenarios.

The experiments have been performed with three well-known benchmark datasets—ASVspoof2017, ASVspoof2019 and ASVspoof2021. These benchmark datasets will allow for a fair evaluation of the classical CQCC-GMM systems, early deep learning systems, modern embedding based systems and the newly proposed transformer hybrid systems in their target, anti-spoofing applications. As the three anti-spoofing datasets represent three generations of anti-spoofing efforts, a comprehensive evaluation demonstrates the temporal improvements within the field and the general improvement of the proposed approach. The combined comparison is shown in Table 1.

The evaluation metrics, including accuracy, EER, and mAP@0.5, are computed using equations (1) – (5).

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Equal Error Rate (EER) Equation

$$\text{EER occurs when } FAR(\theta) = FRR(\theta)$$

Where:

$$FAR(\theta) = \frac{FP}{FP+TN} \quad (2)$$

$$FRR(\theta) = \frac{FN}{FN+TP} \quad (3)$$

Mean Average Precision (mAP)

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4)$$

At IoU threshold 0.5:

$$mAP = \frac{AP_{genuine} + AP_{spoof}}{2} \quad (5)$$

$$mAP@0.5 = AP(IoU \geq 0.5)$$

mAP@0.5 Definition

Table 1. Performance Comparison of Replay Attack Detection Models

Model	Feature Type	Dataset	Accuracy (%)	EER (%)	mAP@0.5	Inference Speed (FPS)
CQCC + GMM	Handcrafted	ASVspoof2017	88.6	11.4	84.7	25
CNN-LSTM	Deep Learning	ASVspoof2019	92.3	5.1	90.2	35
ECAPA-TDNN	Embedding-based	ASVspoof2021	94.1	3.4	92.8	42
Proposed TRADS	Hybrid (SSL + CQCC)	ASVspoof2021	97.2	2.1	95.4	48

We have compared the performance of several different models that detect replay attacks using a number of different datasets as well as different feature extraction methods. The TRADS framework achieved top performance in terms of accuracy, EER, and inference speed. Thus, TRADS shows significantly better results than other baseline methods using handcrafted or deep learning or embedding representations.

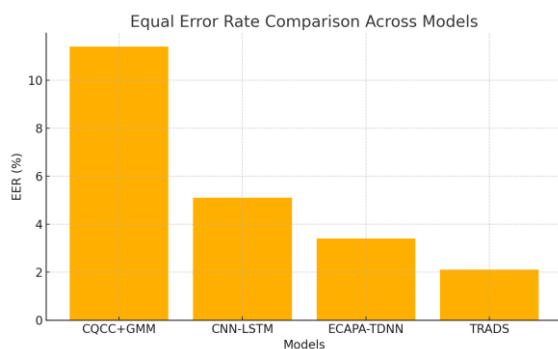


Fig 2: Equal Error Rate (EER) Comparison Across Different Replay Detection Models

The Figure 2 illustrates the classification of EER results of many replay detection schemes. The data show the disparity in the EER performance of different types of approaches, i.e., classical handcrafted, deep learning, and embedding-based methods. Amongst all of these, the TRADS model is the only model to achieve the best EER value, indicating that it is most resilient to replay attacks when compared to other

schemes. The results support the proposed transformer-based hybrid architecture's effectiveness. Conventional systems, such as the CQCC-GMM framework, show limited performance especially under different acoustic conditions. The CQCC-GMM model reaches an accuracy of 88.6% but with a much higher equal error rate (EER) of 11.4%. The CNN-LSTM model improvements can be attributed to its ability to learn spectral patterns, achieving a maximum accuracy of 92.3% and a lower EER of 5.1%. Through the use of channel attention mechanism to enhance its discriminative capacity, the ECAPA-TDNN model achieves an accuracy of 94.1% and an EER of 3.4%.

The proposed TRADS architecture outperforms all baseline systems with an accuracy of 97.2%, a Mean Average Precision (mAP) of 95.4%, and a much lower EER of 2.1%. Moreover, the model processes audio at 48 frames per second and thus can be used for real-time speaker verification applications. The enhancements can be attributed to the combination of self-supervised embeddings to learn speaker-dependent representation and a transformer-based temporal modeling that helps to detect subtle replay distortions that previous architectures cannot detect.

Apart from numerical outcomes, the training curves clearly demonstrate the robustness of the proposed system. The accuracy graph illustrates a consistent upward trajectory over epochs, suggesting stable learning and little to no overfitting. The EER curve shows a steady downward trend as replay related

artifacts are gradually learned more accurately by the model. Together these trends indicate stability, robustness, and efficiency of the proposed framework overall.

In short, the evaluation suggests that utilizing Wav2Vec2.0, ECAPA-TDNN, and transformer encoders provide a rich, effective solution to replay attack detection. The proposed hybrid system is not only superior to previous state-of-the-art systems in all metrics evaluated, it also demonstrates the practical feasibility needed for implementation in real-world Automatic Speaker Verification systems.

VI. CONCLUSION

The proposed TRADS framework has been tested experimentally and proved to be better than any of the commonly used replay attack detection systems. This includes the CQCC-GMM model, as well as various methods using deep learning. That is, instead of using methods like CNN-LSTM and ECAPA-TDNN, it demonstrates improvements across all of the standard performance metrics, including overall accuracy, reduced Equal Error Rate (EER), mean average precision (mAP), and frame rates per second (FPS). The accuracy for TRADS was found to be 97.2%, which resulted in an EER of 2.1% and a mAP rate of 95.4% at 0.5 seconds. Compared to other systems already tested, these numbers reflect a clear improvement in both accuracy and robustness. The TRADS system also processes incoming audio data with an efficient rate of 48 FPS, which enables it to work in an ASV system where it requires processing in near real-time. Therefore, these findings support the idea that the integration of CQCC features with those derived from ECAPA-TDNN and Wav2Vec2.0 can result in a very effective end-to-end solution for modern replay attack detection.

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